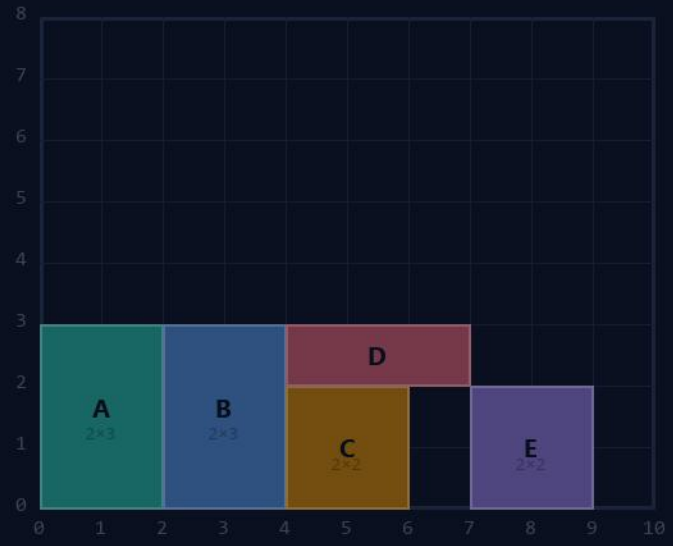
FLOORPLANNING FOR EDA · LAB 08

# Soft module aspect sizing

One module changes shape.

The whole packing has to answer.

`area locked at 6` `aspect 0.5 – 2`




example run · soft packing, A reshaped 2×3

same modules, new neighbors for A

WHY IT MATTERS

# Where this is used

the same loop, on a real chip with hundreds of blocks

## ONE CHIP, TWO KINDS OF BLOCK




hard: vendor fixes w×h · soft: gate count fixes area, rows flex the shape

production tools reshape the soft ones to close gaps before routing

## WHY IT PAYS

**die area**
a few percent is money on every wafer

**wires**
shorter routes → better timing

**power**
shorter routes → less switching energy

**scale**
hundreds of blocks → a search replaces your hand

---

the loop never changes: reshape, place, judge

THE SITUATION

# 23 units of modules, 80 units of outline

a fixed frame, five blocks, one of them free to stretch

## THE OUTLINE




fixed rectangle 10 × 8 → area 80

## THE MODULES

| | | | |
|---|---|---|---|
| A | A 3×2 | area 6 | soft |
| B | B 2×3 | area 6 | hard |
| C | C 2×2 | area 4 | hard |
| D | D 3×1 | area 3 | hard |
| E | E 2×2 | area 4 | hard |

areas 6+6+4+3+4 = 23

## A IS THE SOFT ONE

area locked at 6 — width over height may stretch between 0.5 and 2

B, C, D, E are hard: width and height never change

THE METHOD

# Reshape, place, re-pack, judge

a short loop where the order does the work

THE LOOP

```
1  reshape A:  w·h = 6,  aspect in [0.5, 2]

2  place A into the outline

3  re-pack B, C, D, E around it

4  judge: legal?  what did the numbers do?
```

**step 1 — reshape A**
same area, new w×h

**step 2 — place A**
lower-left corner

**step 3 — re-pack neighbors**
the other half of reshape

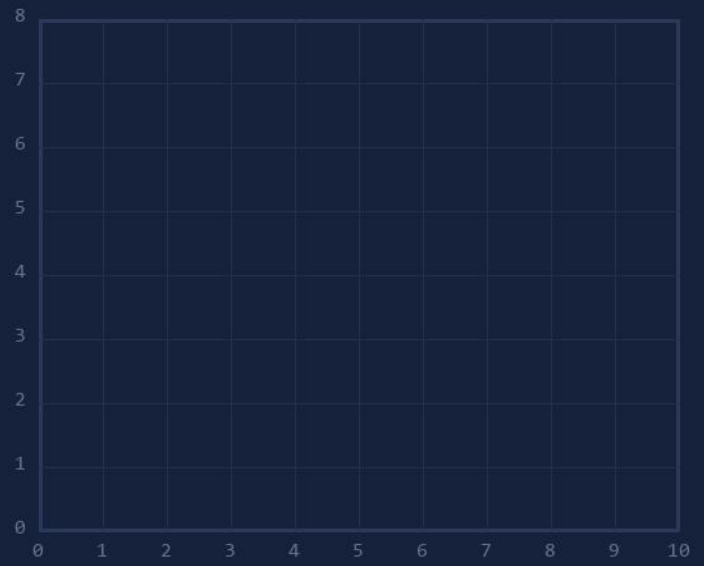
**step 4 — judge**
legal + four numbers

Reshaping is only half the job — the re-pack decides whether the layout is legal.

THE INSTANCE

# Five modules, tiny numbers

small enough to check every claim by hand




example run

MODULE TABLE

| id | w×h | area | aspect bounds |
|---|---|---|---|
| A | 3×2 | 6 | 0.5 – 2 (soft) |
| B | 2×3 | 6 | – |
| C | 2×2 | 4 | – |
| D | 3×1 | 3 | – |
| E | 2×2 | 4 | – |

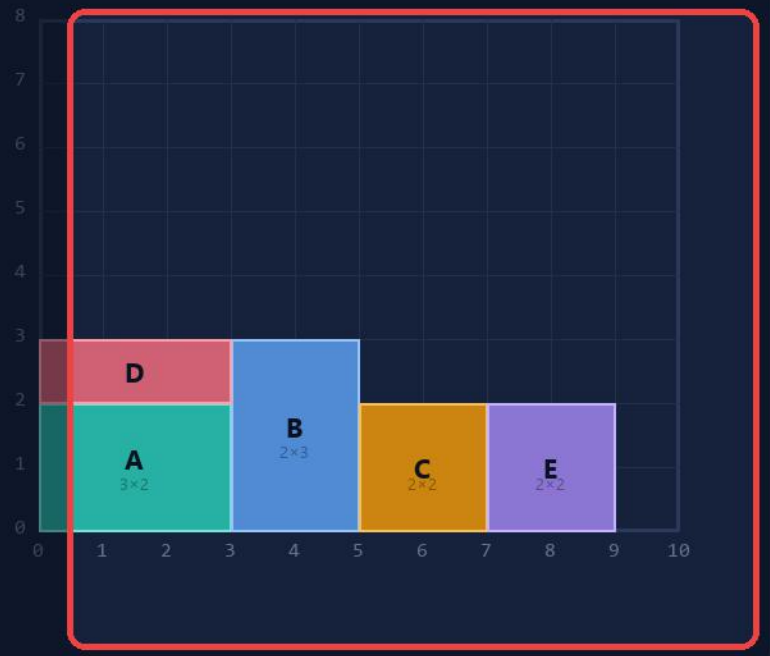
Σ area = 6+6+4+3+4 = 23

outline 10 × 8 = 80 — the numbers are tiny on purpose

BASELINE

# Hard A at 3×2 — the golden packing

every coordinate readable, every cell countable




GOLDEN POSITIONS (lower-left)

A (0,0) 3×2

B (3,0) 2×3

C (5,0) 2×2

D (0,2) 3×1

E (7,0) 2×2

legal: true — inside 10×8, no overlaps

THE ACCOUNTING

`moduleArea 23  outlineArea 80`

deadspace 80 − 23 = 57

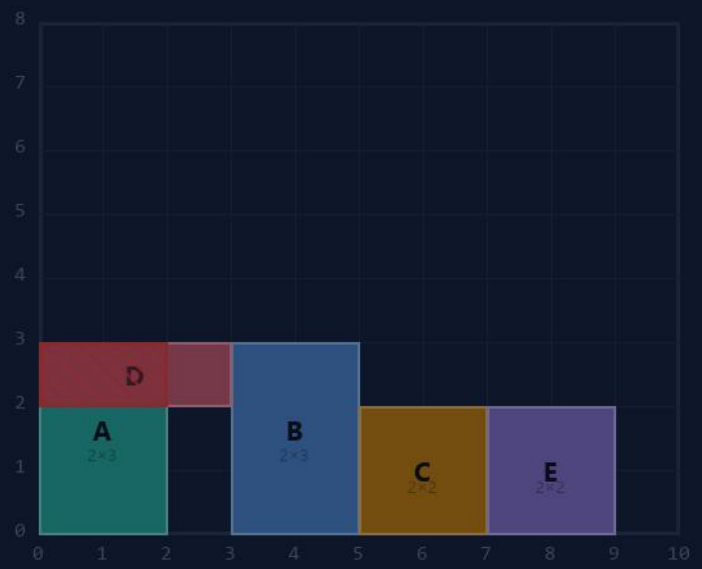
density 23 / 80 = 0.2875 = 28.75%

they follow from the areas alone — not from A's shape

WHAT BREAKS FIRST

# The lazy reshape: A overlaps D

keep A's corner, stretch the shape, watch the checker




example_run

THE ATTEMPT

A stays at (0,0), stretched to 2×3

```
area check: 2 × 3 = 6  −  area kept
```

but D sits at (0,2) — and A now fills that strip too

legal: false

reason: A overlaps D

THE LESSON

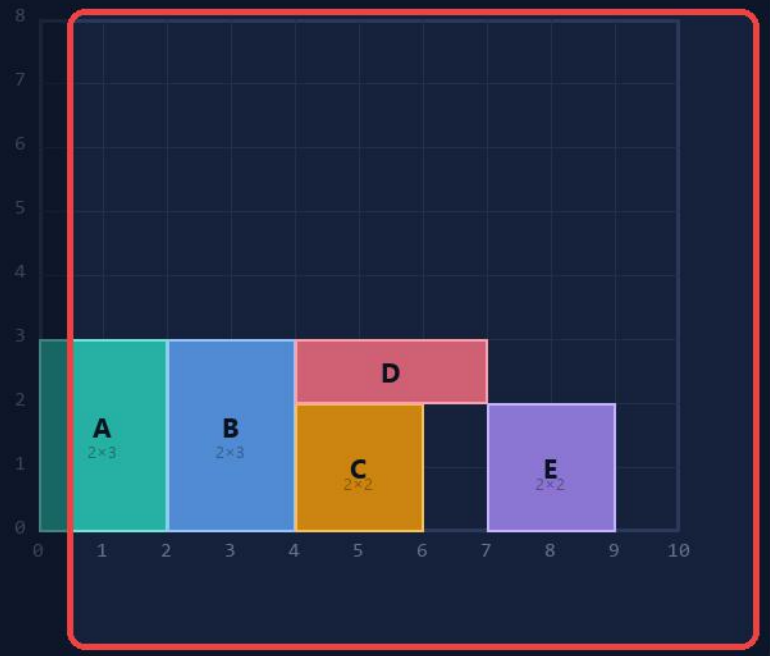
A reshape invalidates neighbors. Re-packing is not optional — it is the other half of reshape.

hatched cells: A and D claiming the same cells

THE LEGAL ANSWER

# Soft golden: A at 2×3, neighbors re-packed

same budget, different floor plan




NEW POSITIONS (lower-left)

legal: true

B (2,0) 2×3

C (4,0) 2×2

D (4,2) 3×1

E (7,0) 2×2

A (0,0) 2×3 reshaped

UNCHANGED TOTALS

deadspace still 57 · density still 28.75%

areas did not move, so the totals cannot move

WHAT MOVED IS THE SHAPE OF THE EMPTY CELLS

hard: pocket of 5 cells above C and E

soft: gap of 2 between C–E (column 6) + strip of 3 beside D

HANDS-ON

# Read the panel like a judge

five lines, five different questions · example run

**soft-module-sizing · example run**

metrics update on every placement

moduleArea: 23

outlineArea: 80

deadspace: 57

density: 0.2875

legal: true

reason: ok

placed: A, B, C, D, E · view: your packing

### moduleArea

the five modules as currently shaped — A's shape is in there

### deadspace

outlineArea − moduleArea = 80 − 23 = 57, reshaping moves it, never shrinks it

### legal + reason

true/false plus the first violation named: "A overlaps D", "E outside outline"

your numbers disagree? read the reason line first

THE FENCE

# Aspect bounds: 0.5 – 2

aspect = width ÷ height — the freedom has edges






1×6

aspect 0.167 ILLEGAL



2×3

aspect 0.667 legal



3×2

aspect 1.5 legal



6×1

aspect 6.0 ILLEGAL

WHY ONLY TWO RESHAPE BUTTONS

with area 6 on an integer grid, exactly 2×3 and 3×2 stay inside the fence

and density is area ÷ area — no shape of A lifts 28.75%: softness rearranges whitespace, it never creates area

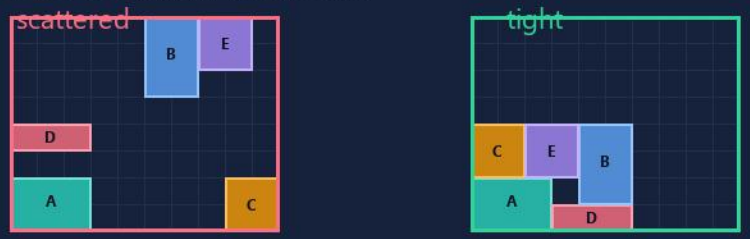
THE ONE TRAP

# Legal is not good

the checker answers a yes or no question, not a quality question

## TWO LEGAL PACKINGS, SAME MODULES

A B C D E placed inside 10×8




bbox 10×8 = 80
legal: true




bbox 6×4 = 24
legal: true

both pass the checker; only one uses the reshape

## WHAT LEGAL ACTUALLY CHECKS

inside the frame

no positive-area overlap

— and nothing else

## SO FIVE SCATTERED MODULES PASS

But the bounding box is still the whole frame, and the reshape bought you nothing.

## READ THE BBOX NEXT TO LEGAL

Legality is a gate, not a score. A packing can be legal and still be a bad floorplan.

this is what challenge 2 asks for: bbox 24

YOUR TURN

# Reshape it, pack it, judge it

the lab opens from the section under this clip

YOUR CHECKLIST

1. Warm up: reshape A to 2×3 — area stays 6.
2. Main task: place all five until legal is true.
3. Any legal packing counts — golden not required.
4. Challenge: pin D at (0,0); bbox ≤ 24, one wasted cell.
5. Watch area · density · deadspace · legal as you place.

Reshape A | Check | Hint

NUMBERS TO WATCH

`deadspace 57 · density 28.75%`

they never move — only the shape does

NEXT

Fix macro D at the origin and pack around it.

then

the quiz

then the next lab